Programming Project 3: OOP Library Management

Start Assignment

- Due Thursday by 11:59pm
- Points 100
- Submitting a text entry box or a file upload
- File Types cc and h
- Available until Mar 22 at 11:59pm

Classes to be supported:

- Book Class
  - Methods
    - Book(string t, string a, int id)
    - Book(const Book& other)
    - ~Book()
    - checkout()
    - returnBook()
    - bool isAvailable()
    - string getTitle()
    - int getId()
    - bool operator== Books equal when id matches
  - Data members
    - string title
    - string author
    - int id
    - bool checkedOut
- User Class
  - Methods
    - borrow(book & b)
    - return Book& b
    - overload <<
  - Data Members
    - string name;
    - vector<Book *>borrowedBooks;
- Library Class:
  - Methods:
    - addBook(YOUDEF)
    - addUser(YOUDEF)
    - Book * findBook(id);

        - int getBookIdByTitle(string)
        - int getUserIdByName(string)
        - checkoutBook(int userid, int bookid);
        - returnBook(int user id, int bookid);
        - ovearloaded << operator that implements print inventory
    - Data members:
        - vector<Book> inventory;
        - vector<User> users;

Sample main content:

```
int main() {

Library lib;

lib.addBook("Art of the Deal", "DJ Trump");

lib.addBook("C++", "Deitel");

User u("Patel");

lib.addUser(u);

lib.checkout(lib.getUserIdByname("Patel"), lib.getBookIdByTitle("Art of the Deal"));

cout << u; // print list of books borrowed by user

cout << lib; // print inventory of books indicating those that are out

}
```

}

Project 3

Criteria	Ratings		Pts
Compilation Project compiles without any errors or warnings.	**10 pts** **Full marks**	**0 pts** **No marks**	10 pts
Test case 1: Book Class Initialization Input: Book b("The Hobbit", "J.R.R. Tolkien", 101) Output: getTitle() returns "The Hobbit", isAvailable() returns true	**10 pts** **Full marks**	**0 pts** **No marks**	10 pts
Test case 2: Book Operator Overloading Input: Book b1("A", "B", 101), Book b2("C", "D", 101) Output: b1 == b2 returns true (since IDs match)	**10 pts** **Full marks**	**0 pts** **No marks**	10 pts
Test case 3: User Class Borrow Logic Input: User u("Patel"), u.borrow(b1) Output: Book b1 is successfully added to the user's borrowedBooks vector	**10 pts** **Full marks**	**0 pts** **No marks**	10 pts
Test case 4: User Operator Overloading Input: cout << u (after Patel borrows a book) Output: Prints user's name ("Patel") and a list of their borrowed books	**10 pts** **Full marks**	**0 pts** **No marks**	10 pts
Test case 5: Library Add & Search Books Input: lib.addBook("Art of the Deal", "DJ Trump"), lib.getBookIdByTitle("Art of the Deal") Output: Returns the valid integer ID assigned to the book	**10 pts** **Full marks**	**0 pts** **No marks**	10 pts
Test case 6: Library Add & Search Users Input: lib.addUser(u), lib.getUserIdByName("Patel") Output: Returns the valid integer ID assigned to the user	**10 pts** **Full marks**	**0 pts** **No marks**	10 pts

Criteria	Ratings		Pts
Test case 7: Library Checkout Transaction Input: lib.checkoutBook(userId, bookId) Output: Book's checkedOut status becomes true, and it is added to the user's list	**10 pts** **Full marks**	**0 pts** **No marks**	10 pts
Test case 8: Library Return Transaction Input: lib.returnBook(userId, bookId) Output: Book's checkedOut status becomes false, and it is removed from the user's list	**10 pts** **Full marks**	**0 pts** **No marks**	10 pts
Test case 9: Library Operator Overloading Input: cout << lib Output: Prints the entire library inventory, clearly indicating which books are out	**10 pts** **Full marks**	**0 pts** **No marks**	10 pts
			Total Points: 100